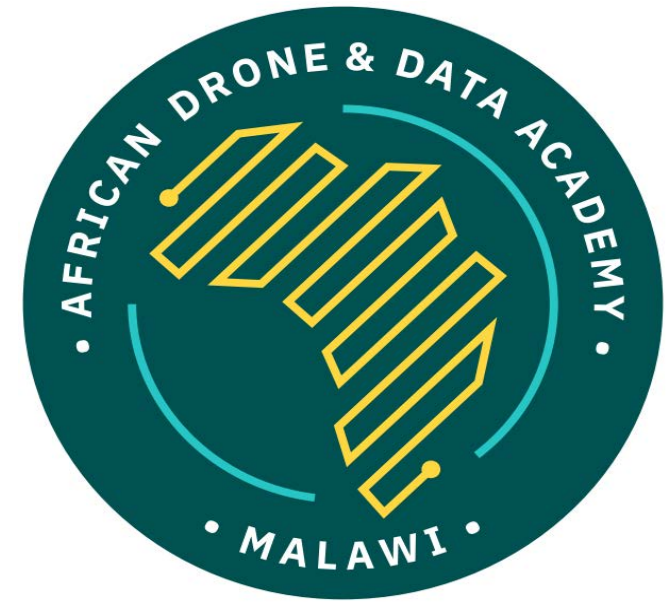




TOP Level 1

Day 8 Version A





TOP Level 1

Meteorology



Meteorology is the study of weather both on a large and small scale

- Weather information is either obtained from government sources or commercial services that provide details about surface conditions (on the ground) and synoptic conditions (atmospheric trends) useful for flight planning
- All available weather data should be accessed in flight planning so there are few surprises encountered during flight
 - A weather forecast can have a strong influence on the exact mission flown, overriding other planning tools used in the process.

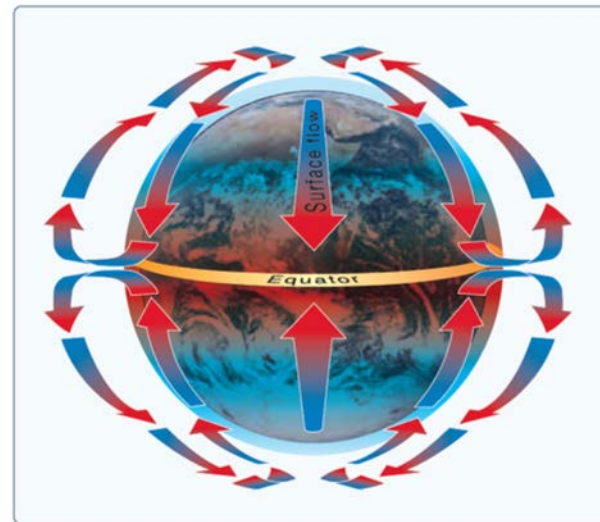
11. RPA operations
12. Flight performance
- 13. Meteorology**
14. Navigation theory
15. Commercial ops
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2



Meteorology is the study of weather both on a large and small scale

- On a large scale, global weather patterns are caused by longitudinal air circulation due to the temperature differences at the poles and at the equator
- These axial air currents in turn cause lateral air currents due to Coriolis acceleration
- The bands of circulation that result govern the weather in zones, although some of this predictability is diminishing with global warming

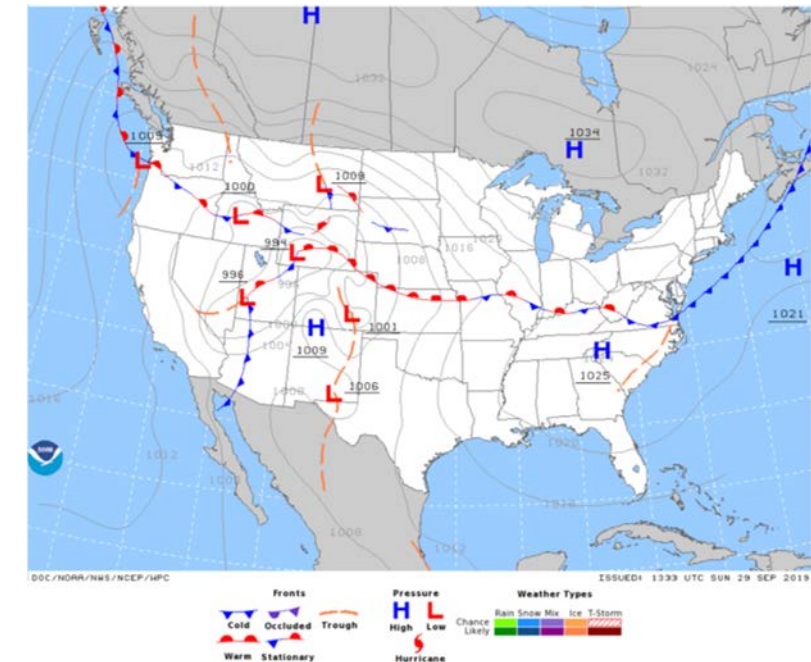
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12. Flight performance
- 13. Meteorology**
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15. Commercial ops
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2



A weather forecast is a critical part of the day-before flight planning

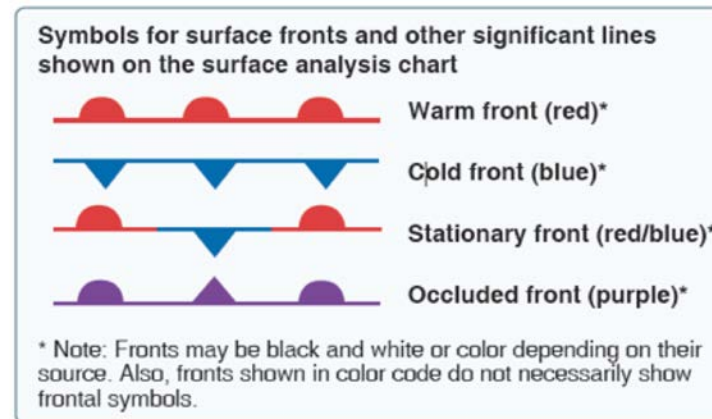
- Forecasts are based on large-scale weather models, shown on surface analysis charts that depicts fronts, high pressure systems and low pressure systems
- These systems define the behavior of winds and the trends that can be expected during the day that will affect flying conditions
- Generally speaking:
 - Fronts bring cloudy conditions
 - High pressure brings favorable conditions
 - Low pressure brings unfavorable conditions

11. RPA operations
12. Flight performance
- 13. Meteorology**
14. Navigation theory
15. Commercial ops
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2

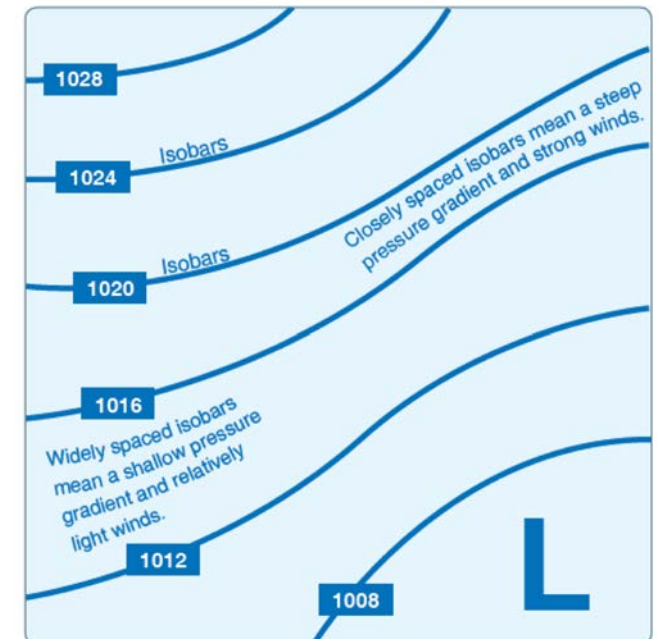


A weather forecast is a critical part of the day-before flight planning

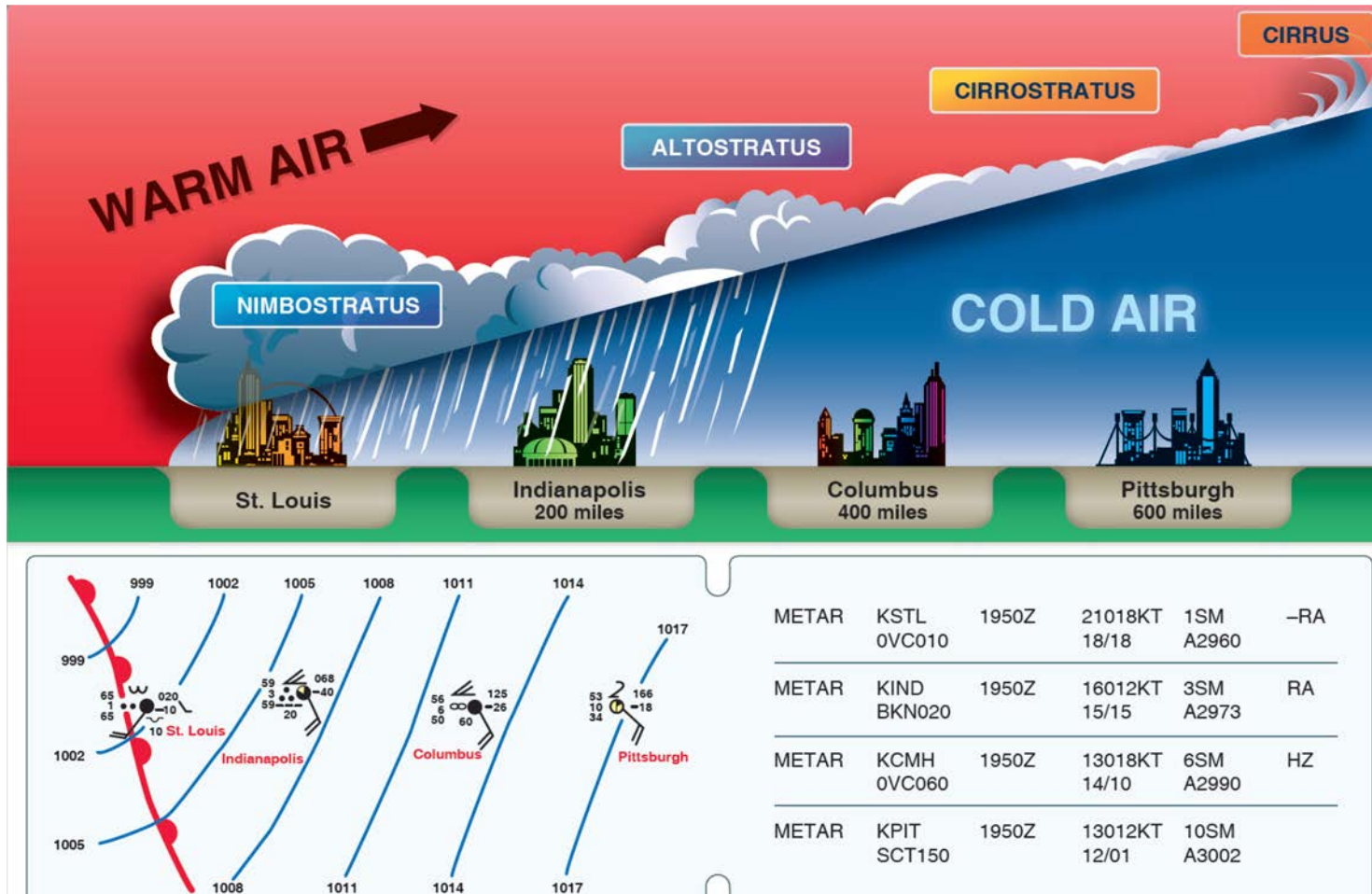
- The surface analysis (SA) chart reveals information about air masses that is helpful in understanding what kind of weather we can expect
- The SA chart shows warm and cold fronts, high and low pressure centers, and isobars that indicate changes in pressure



11. RPA operations
12. Flight performance
- 13. Meteorology**
14. Navigation theory
15. Commercial ops
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2



Warm fronts that move into an area usually bring rain and low visibility

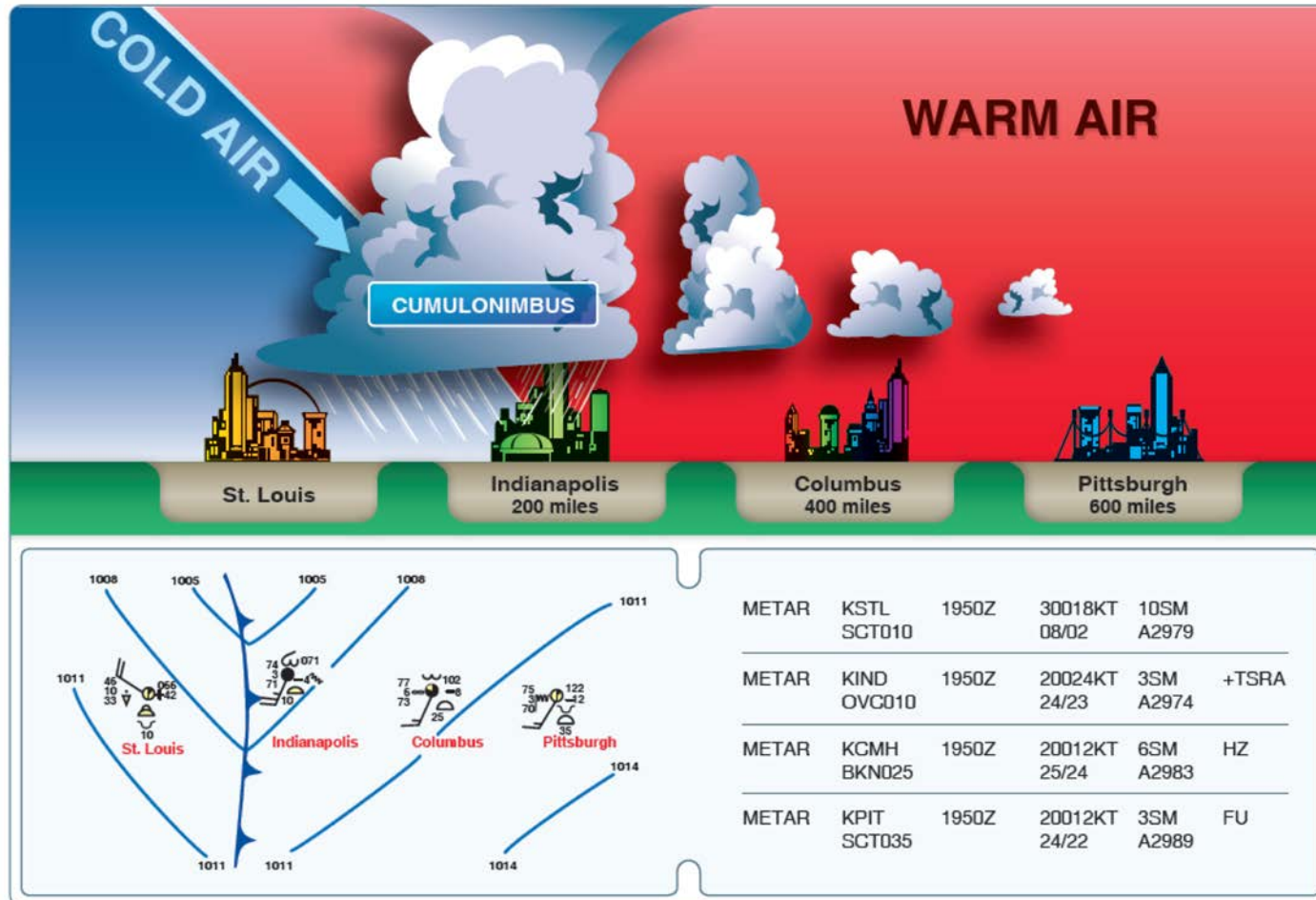


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12. Flight performance
13. **Meteorology**
14. Navigation theory
15. Commercial ops
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2

- A warm front is characterized by a slow moving line of warm air overtaking cooler air
 - A warm front is usually indicated by warm, moist air moving into the region
 - Low visibility is typical

Cold fronts that move into an area usually bring wind and severe weather

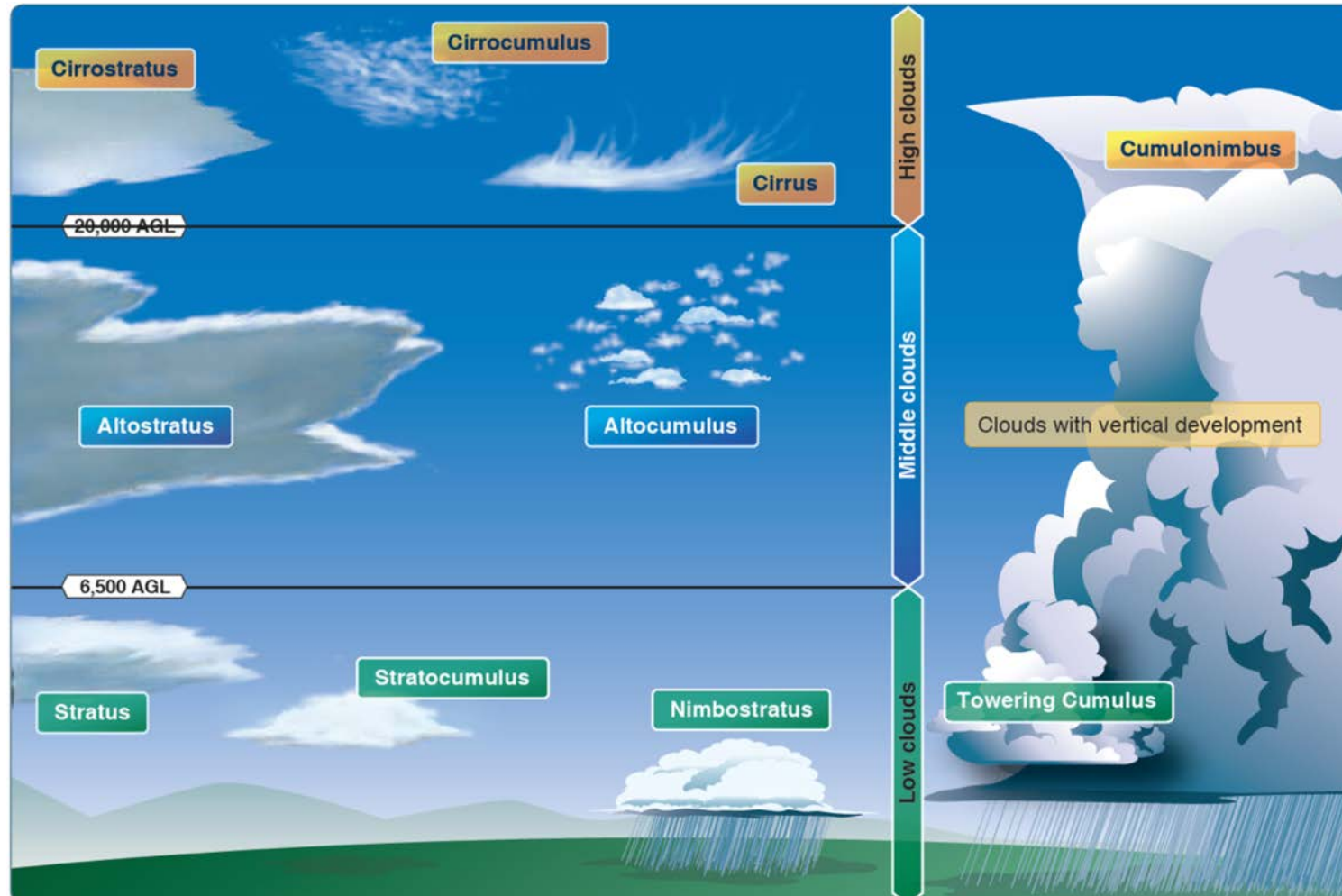
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12. Flight performance
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15. Commercial ops
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2



- A cold front is characterized by a fast moving line of cold air that moves in under warm air
 - Cold fronts are stormy and should be avoided by all types of aircraft

Clouds can indicate the types of flying conditions to be expected

11. RPA operations
12. Flight performance
- 13. Meteorology**
14. Navigation theory
15. Commercial ops
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2



- Severe weather (convective activity) is indicated by cumulonimbus clouds - sometimes with an anvil type
 - Lightning should be expected
 - Strong, gusty winds with rain and hail

A weather forecast is a critical part of the day-before flight planning

- Forecasts are also known as **TAFs** (terminal aerodrome forecast)
- They can be obtained by the South African Weather Service
- The TAF is usually accompanied by an observation report called the **METAR** (meteorological aviation routine report)
- Both the METAR and TAF use a coded system for describing weather conditions

11. RPA operations
12. Flight performance
- 13. Meteorology**
14. Navigation theory
15. Commercial ops
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2

FWKI - Lilongwe International

METAR

FWKI 291600Z 06008KT CAVOK 25/11 Q1016
NOSIG

TAF

FWKI 291500Z 2918/3024 09012KT CAVOK
BECMG 3004/3006 10006KT 8000 HZ SKC
BECMG 3010/3012 08008KT 9999 FEW030
FM301400 08008KT CAVOK

The codes used in METARs and TAFs should be understood since they contain important weather information

- 11. RPA operations
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- 13. **Meteorology**
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- 19. Proficiency 2



Key to Aerodrome Forecast (TAF) and Aviation Routine Weather Report (METAR) (Back)



	In METAR , <u>ReMarK</u> indicator & remarks. For example: <u>Sea- Level Pressure</u> in hectoPascals & tenths, as shown: 1004.5 hPa; <u>Temp/dew-point</u> in tenths °C, as shown: temp. 18.2°C, dew-point 15.9°C	RMK SLP045 T01820159
FM091930	<u>FroM</u> : changes are expected at: 2-digit date, 2-digit hour, and 2-digit minute beginning time: indicates significant change. Each FM starts on a new line, indented 5 spaces	
TEMPO 0920/0922	<u>TEMPO</u> rary: changes expected for <1 hour and in total, < half of the period between the 2-digit date and 2-digit hour beginning, and 2-digit date and 2-digit hour ending time	
PROB30 1004/1007	<u>PROB</u> ability and 2-digit percent (30 or 40): probable condition in the period between the 2-digit date & 2-digit hour beginning time, and the 2-digit date and 2-digit hour ending time	
BECMG 1013/1015	<u>BEC</u> o <u>MinG</u> : change expected in the period between the 2-digit date and 2-digit hour beginning time, and the 2-digit date and 2-digit hour ending time	



The codes used in METARs and TAFs should be understood since they contain important weather information

- RPA operations
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 - Navigation theory
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Qualifier		Weather Phenomena		
Intensity or Proximity 1	Descriptor 2	Precipitation 3	Obscuration 4	Other 5
– Light	MI Shallow	DZ Drizzle	BR Mist	PO Dust/sand whirls
Moderate (no qualifier)	BC Patches	RA Rain	FG Fog	SQ Squalls
+ Heavy	DR Low drifting	SN Snow	FU Smoke	FC Funnel cloud
VC in the vicinity	BL Blowing	SG Snow grains	DU Dust	+FC Tornado or waterspout
	SH Showers	IC Ice crystals (diamond dust)	SA Sand	SS Sandstorm
	TS Thunderstorms	PL Ice pellets	HZ Haze	DS Dust storm
	FZ Freezing	GR Hail	PY Spray	
	PR Partial	GS Small hail or snow pellets	VA Volcanic ash	
		UP *Unknown precipitation		
The weather groups are constructed by considering columns 1–5 in this table in sequence: intensity, followed by descriptor, followed by weather phenomena (e.g., heavy rain showers(s) is coded as +SHRA). * Automated stations only				



Generally speaking, the TAF is used for day before flight planning, and both the TAF and METAR are used the day of the flight

- Winds, visibility and ceiling are the most important information for flight planning
 - Most countries only allow RPAs to fly up to 400 ft. AGL (above ground level), so clouds are usually not a problem
 - The operator must always maintain visual line of sight of the aircraft, meaning that if the visibility is poor, then the flight radius will be small
 - Winds are reported as steady and gusts (02010KTG20). The maximum predicted windspeed (steady or gust) cannot exceed the operational limits of the aircraft

11. RPA operations
12. Flight performance
13. **Meteorology**
14. Navigation theory
15. Commercial ops
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2



Trial METAR interpretation

11. RPA operations
12. Flight performance
- 13. Meteorology**
14. Navigation theory
15. Commercial ops
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2

  **AVIATION WEATHER CENTER**
NOAA NATIONAL WEATHER SERVICE

Local Forecast [HOME](#) [ADVISORIES](#) [FORECASTS](#) [OBSERVATIONS](#) [TOOLS](#) [NEWS](#) [SEA](#)

METAR Data [METAR Home](#) [Plot](#) [Data](#) [Board](#) [Info](#)

IDs: Format: ☐ Raw ☒ Decoded ☐ Include TAF

Data at: 2117 UTC 29 Sep 2019

METAR for: KBCB (Blacksburg/VA Tech A, VA, US)

Text: KBCB 292115Z AUTO 12007KT 10SM SCT065 BKN080 BKN100 29/19 A3025 RMK AO2 T02940190

Temperature: 29.4°C (85°F)

Dewpoint: 19.0°C (66°F) [RH = 54%]

Pressure (altimeter): 30.25 inches Hg (1024.5 mb)

Winds: from the ESE (120 degrees) at 8 MPH (7 knots; 3.6 m/s)

Visibility: 10 or more sm (16+ km)

Ceiling: 8000 feet AGL

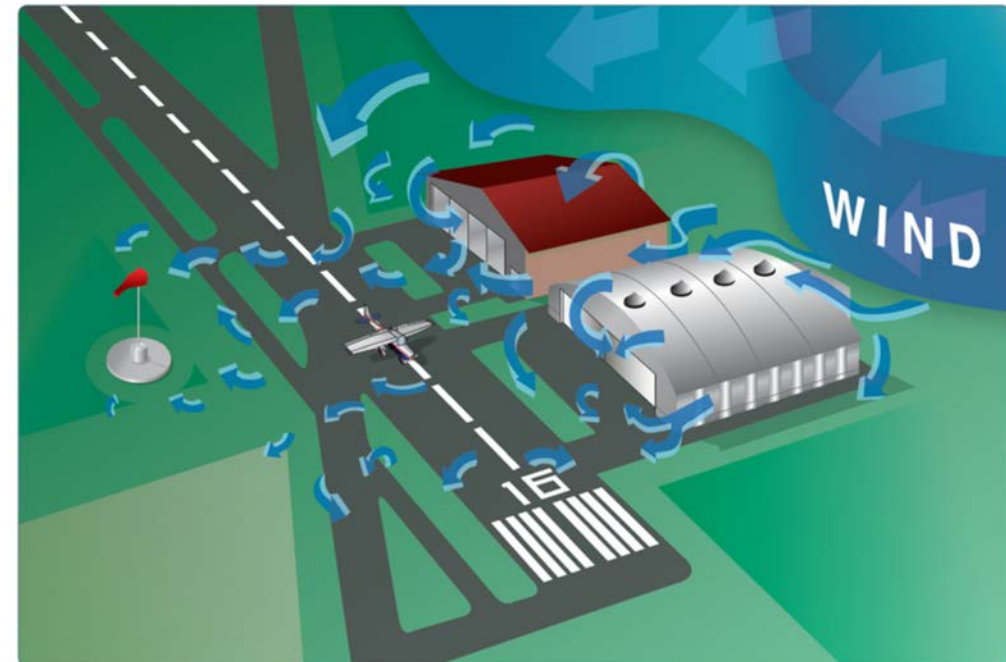
Clouds: scattered clouds at 6500 feet AGL, broken clouds at 8000 feet AGL, broken clouds at 10000 feet AGL

QC Flag: automated observation with no human augmentation

Large scale winds will have an effect at lower altitudes where winds interact with terrain features

- Any time flights are taking place downwind of structures that can cause the airflow to be disturbed, expect turbulence
- Turbulent conditions - gusty and unpredictable winds - will occur whenever there is a breeze around buildings and trees
- It is always best to choose a flight location that has a relatively smooth terrain profile upwind of the flying site

11. RPA operations
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16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2



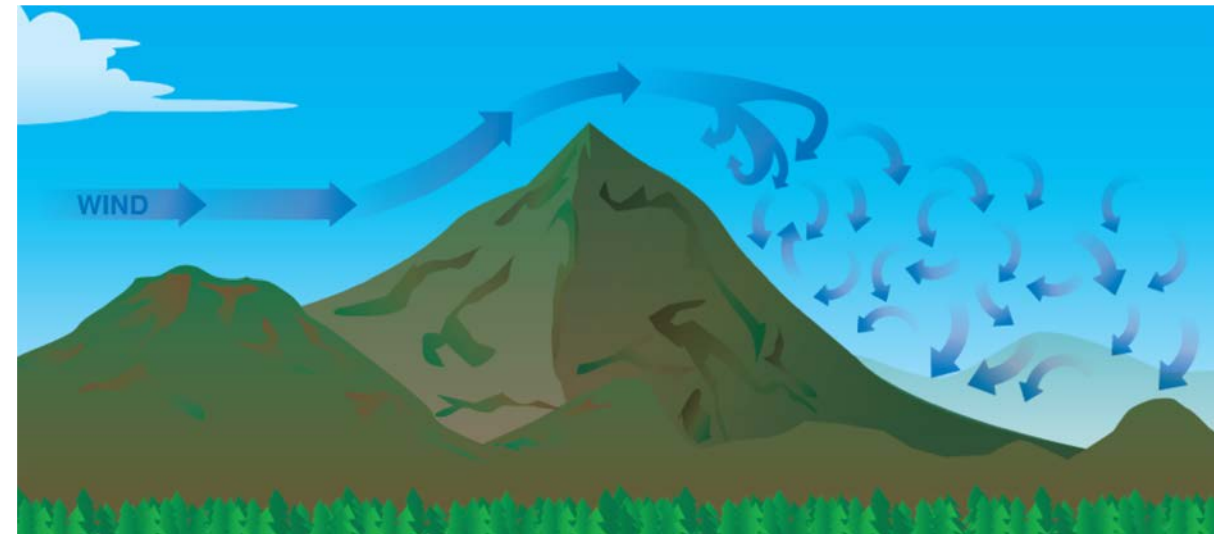
Large scale winds will have an effect at lower altitudes where winds interact with terrain features

- Downwind of mountains and hills can be particularly challenging areas to fly in since the a strong downwash can be experienced, forcing the aircraft down

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15. Commercial ops
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2



Altitude plot of Gogode - Kasungu flight Nov 2017



How is the aircraft effected by the weather?

11. RPA operations
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15. Commercial ops
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17. Human performance
18. Proficiency 1
19. Proficiency 2

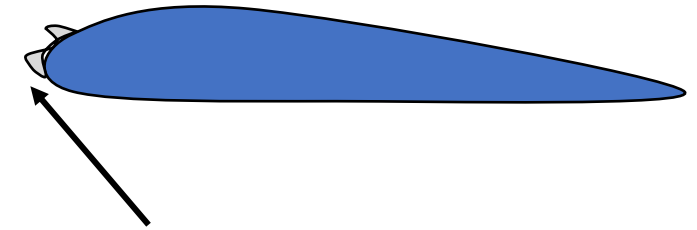
Weather condition	Effect on aircraft	Effect on battery life	Effect on performance	Effect on controllability
Wind	Causes flight time to increase or decrease which can impact the mission	Not a direct affect unless the majority of the flight is in terrain-induced downwash	Mission may not be completed if there is a strong headwind component	Aircraft should be controllable unless turbulence exceeds the maneuvering capability of the aircraft
Precipitation	Many drones are not capable of flying in visible precipitation without damage to the electronics	Precipitation will cause more drag in the aircraft which will accelerate power consumption	If the aircraft can fly in visible precipitation, the performance will be degraded due to additional drag	The controllability of the aircraft should not be strongly impacted by precipitation
Altitude	None	None	Significant degradation of performance due to higher density altitude which reduces lifting capacity	None
Temperature	None	Batteries will have degraded performance above 40°C and below 10°C	A lower temperature results in a lower density altitude which increase aerodynamic performance	None



How is the aircraft effected by the weather?

- **Airframe or propeller icing** can occur on any aircraft when flying in visible moisture (clouds) or precipitation when at or below the freezing temperature
 - An **icing condition is dangerous** because it strongly **degrades the lift** generated by the wings or propellers, and it **increases drag** which **increases power consumption**
 - An increase in power consumption will in turn reduce the flight endurance
- Icing conditions, if they exist, are usually noted in a weather briefing
 - This could be in the form of a pilot report (someone who experienced icing first hand), or in a forecast product because the temperature and moisture conditions are ideal for ice formation

11. RPA operations
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Ice can form an irregular shape on the leading edge of the wing, greatly impacting the ability of the wing to generate lift while increasing the drag



TOP Level 1 Proficiency



Pilot proficiency

- Pilot skill and proficiency is not “one and done,” but requires constant attention to maintain a high performance level
 - Pilots who are flying frequently will retain more “muscle memory” on flight control and decision-making under stressful conditions
 - Either a pilot flies frequently in their training or job, or they must log proficiency flights at a regular interval to stay current
- *Pilot proficiency is equally important for RPIC and manned aircraft pilots*

11. RPA operations
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- 18. Proficiency 1**
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Pilot proficiency

- FOM Sec 3.5.4.6 outlines the terms for flight currency:
 - To be considered current, an RPIC must perform every 30 consecutive days, one launch and recovery and 15 minutes of flight time for each type of RPA qualified to operate (e.g. multirotor or fixed-wing)
 - Any crewmember whose currency has lapsed for more than 30 days will enter a 30-day probationary period in which they must accomplish two launch and recoveries and 30 minutes of flight in each type of RPA qualified to operate. RPICs who fail to complete the probationary requirements or allow more than 60 days since their currency had lapsed will enter suspension and must complete a proficiency flight evaluation that is consistent with the TOP Level 2 maneuvers.

11. RPA operations
12. Flight performance
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16. Principles of flight
17. Human performance
- 18. Proficiency 1**
19. Proficiency 2

Pilot proficiency

- In order to maintain pilot proficiency, the ADDATracking system has been developed that will allow pilots to monitor their currency and respond to lapses if needed

11. RPA operations
12. Flight performance
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19. **Proficiency 2**

Pilot Logbook					Enter the ID of the Pilot Logbook you wish to see								
					190102								
Pilot ID	Total Flight Time (minutes)	Lesson 1	Lesson 1 Pass Date	Lesson 1 Notes	Lesson 1 Practice Recommendations	Lesson 2	Lesson 2 Pass Date	Lesson 2 Notes	Lesson 2 Practice Recommendations	Lesson 3	Lesson 3 Pass Date	Lesson 3 Notes	Lesson 3 Practice Recommendations
190102	18	Pass	11/18/2019	Proficient. Impressive performance.	Should practice landing more softly, if anything.	Pass	11/18/2019	passed quickly, good job	work on movement in two directions simultaneously				

- The logs show total flight time, lessons passed, date for each pass, notes, and recommendations for improvement



TOP Level 1 Compliance



Inspection and maintenance programs

- Aircraft inspections include both **generic items** for a class of aircraft and **specific items** that are indicated in the OEM user manual
 - Inspection procedures for the S-500 are documented in the S-500 user manual and are duplicated in the FP01 checklist in *ADDATracking*
 - *Many inspection items that apply to the S-500 specifically are also applicable for a wide class of small VTOL drones*
 - In using *ADDATracking*, **inspection items are sequentially presented to the RPIC and require completion confirmation to proceed to flight**

- 20. Compliance
- 21. Documentation
- 22. Aircraft logs
- 23. Maintenance logs
- 24. Battery logs



Inspection and maintenance programs

- Generic maintenance classifications and procedures are outlined in the General Maintenance Manual (GMM)
 - This manual designates the authority to conduct maintenance and defines the documentation of maintenance activity
- Specific maintenance for an aircraft is described in the user manual
- The flight operations manual (FOM) Sec 7.4 guides maintenance practice at the ADDA
 - In addition to airframe maintenance, **battery logbooks** and **software updates** are also included in a maintenance program
 - **Logged maintenance is managed in *ADDATracking***

- 20. Compliance
- 21. Documentation
- 22. Aircraft logs
- 23. Maintenance logs
- 24. Battery logs





TOP Level 1

Logs



Accessing aircraft and component logbooks

- 20. Compliance
- 21. Documentation
- 22. Aircraft logs**
- 23. Maintenance logs**
- 24. Battery logs**

- The aircraft logbooks are accessible via shortcut on the *ADDATracking* devices, similar to user manual and reference document access
 - **Flight data entered** into *ADDATracking* through the forms system allows **generation of accessible reports** for the following:
 - Pilot logbooks (restricted access)
 - Aircraft logbooks (common access)
 - Maintenance logs, to include batteries and software update information (common access)
 - Certificates, waivers and approvals



Accident and incident reporting is performed to track, learn and improve

- Tracking of hazards, accidents and incidents is completed through the *ADDATracking* system and will allow weaknesses in the ADDA program to be identified
 - The chief pilot and aviation safety officer are notified through *ADDATracking* when an event occurs
 - They are given the opportunity to determine an outcome of the event, such as recurrent training, changes to the program, or disciplinary action taken against the pilot
 - Events that are not considered “typical” (such as a prop strike) will be addressed by the internal safety review committee
- The logging system allows hazards (pre-accident or incident) to be identified both by operators of the aircraft and also by outside observers through the Anonymous Safety Reporting system

31. Accident records
32. Accident reporting
33. Safety training

Incidents

Any deviation from the planned flight should be recorded in the flight log with no exceptions.

Check any incidents that occurred during the flights

☐ NONE	☐ Injury	☐ Death
☐ Property Damage	☐ Prop Strike/Tipover	☐ Hard Landing
☐ Foreign Object Damage	☐ Wildlife	☐ Airspace Violations
☐ Traffic Conflict in Flight	☐ Near Midair Collision	☐ Procedure Error
☐ Human Factor Error	☐ Fatigue	☐ Fuel/Energy Event
☐ Altitude Deviations	☐ Loss of Data Downlink	☐ Loss of Command Uplink
☐ Loss of Telemetry Downlink	☐ Other (explain clearly)	

Check all that apply. An incident report is required for any response other than NONE.

2 pages left

Next Page



TOP Level 1

Personal Protective Equipment



PPE equipment - specified in the SMS

- Personal Protective Equipment (PPE) is used to reduce the risk to the crew related to UAS-specific hazards and is required for all operations of UAS with the ADDA program. The minimum requirements are:
 - **Safety glasses**
 - **Sleeved shirt (short or long)**
 - **Closed-toe shoes**
- Additional PPE may be required based on the risk assessments/mitigation plans for specific mission plans.
- **ADDA will provide safety glasses to personnel and students that do not have them.** Personnel and students are responsible for having appropriate clothing and shoes.

25. Safety policy
26. **PPE equipment**
27. **PPE access**
28. Job safety check
29. Log of hazards
30. Emerg resp plan





TOP Level 1

Job Safety Check



Job safety assessment - every flight

- In order to reduce the likelihood of injury and accident, the ADDA has several checks for safe flight operation
 - **FOM Sec 3.4: Risk Assessment** specifies a procedure which is implemented in *ADDATracking* to check the overall risk of the mission, and proceed to flight operations if the risk is below a threshold
 - **The SMS** more broadly discusses the promotion of a safety culture within the ADDA
 - **Before every flight, a standard briefing AC01** is conducted to verify that all crew members are **wearing necessary PPE** for the mission

25. Safety policy
26. PPE equipment
27. PPE access
- 28. Job safety check**
29. Log of hazards
30. Emerg resp plan





TOP Level 1

Log of Hazards



Hazard identification and documentation

- Potential hazards may be observed, but unless there is an accessible reporting system available, they may not be documented
- The ADDA maintains an open and anonymous safety reporting system (ASR) that anyone can use, even non-ADDA members
 - The ASR is used to report any safety concerns observed in flight operations
 - All submissions are equally investigated, assessed, and addressed immediately upon receipt by the Safety Committee
 - The ASR form is confidential, and information about how to access it is provided on signage that is displayed on the ADDA premises

- 25. Safety policy
- 26. PPE equipment
- 27. PPE access
- 28. Job safety check
- 29. Log of hazards**
- 30. Emerg resp plan